# HYPOTHESIS TESTING

By Sharvari Deshmane

# Hypothesis testing

- Level of Significance
- Null and alternate hypothesis
- Types of error
- *Operating Characteristic* (or OC)
- Power of test
- Test Statistics
- p-value
- Types of test

# Hypothesis testing

- **Testing for mean of one normal population**
  - Known population variance
  - Unknown population variance: t test
- **Testing the equality of means of two normal populations**
  - Case of Known Variances
  - Case of Unknown and equal Variances

# Parameter Estimation

- Whenever a population contains unknown parameters, such parameters can be estimated either with a single value(point estimator) or two real values within which the parameter lies(interval estimation)
- These estimates are based on a single sample($x_1$,$x_2$,….,$x_n$) of size n

# Need of Testing of Hypothesis

- Assume a situation in which we have sample in our hand whose population is not known but suspect to be from a population with parameter $\theta$

# Need of Testing of Hypothesis

- To verify whether the sample drawn is from a population with parameter θ we follow:
  - Obtain sample statistics related to θ (i.e. if θ is mean of population then the statistics must also be mean of the sample)
  - Perform statistical test to know whether the sample drawn is from the population with parameter θ
  - If the test is in favour of the sample, then we conclude that the sample belongs to the population otherwise it comes from some other population

# Testing of Hypothesis

- The process of assuming a population to the unknown sample and testing for its truth i.e. whether the sample comes from the population or not is known as testing of Hypothesis

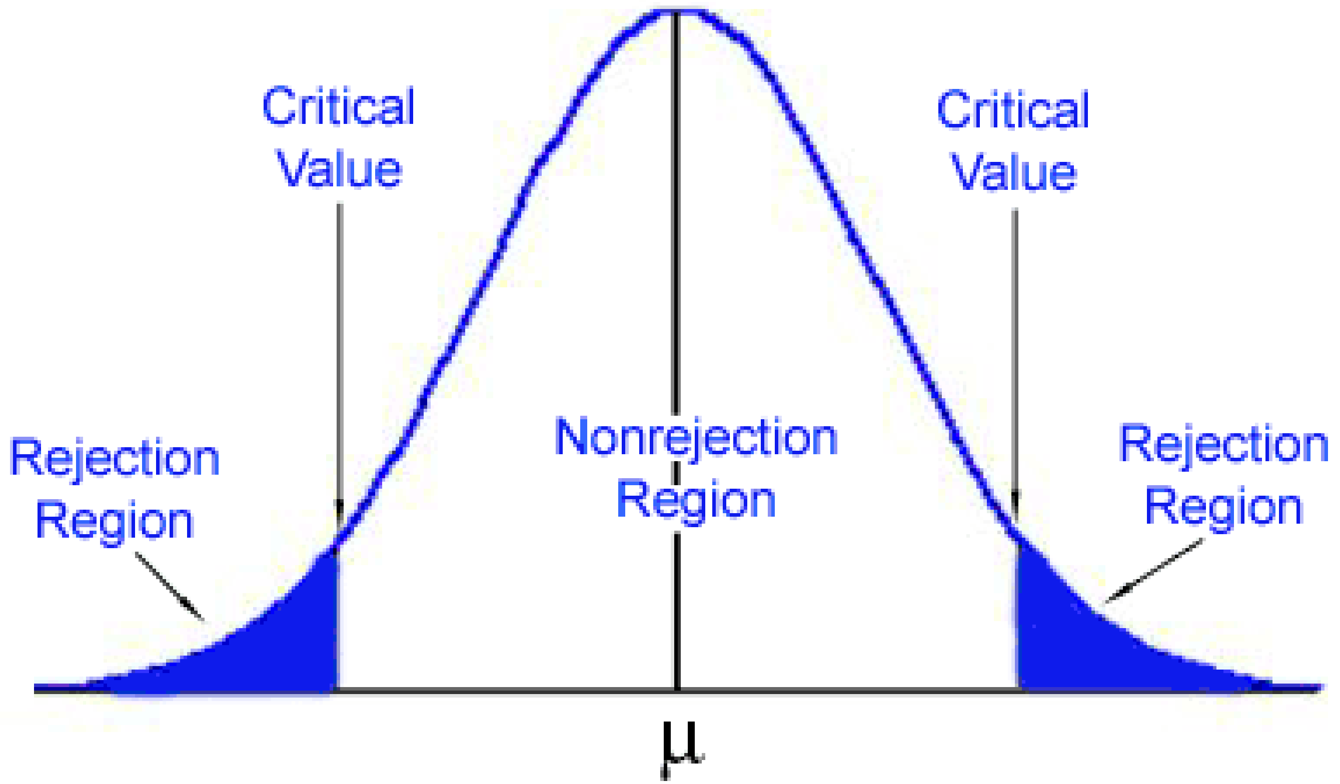
# Level of Significance

- The decision taken whether the sample belong to the population or not may involve some sort of error
- The error is due to the fact that the different samples drawn from the same population can have different sample statistics and hence result in different conclusions.
- Hence the experimenter fix the limit for the probability of committing such an error

# Level of Significance

- This prefix level at which the sample statics may differ or do not differ from assumed population parameter is known as *level of significance*
- It is similar to that of the confidence coefficient in which while (1-$\alpha$) gives the probability that the population parameter lies within the interval determined
- (1-$\alpha$) is acceptance region whereas $\alpha$ which is rejection area is the level of significance

# Level of Significance



# Null and alternate hypothesis

- Assume a situation where we have sample ($x_1$, $x_2$,…..,$x_n$) of size n but we do not know from which population this sample has been drawn
- Let us assume the sample has been drawn for the population with parameter θ
- The testing of hypothesis deals with making assumptions about the unknown parameter θ, say $\theta = \theta_0$, of the population from where the sample has come

# Null and alternate hypothesis

- When we test a hypothesis,
- The test is true :

  *Null hypothesis* $H_0: \theta = \theta_0$

- The test is not in favour of $H_0$:

  *Alternate Hypothesis* $H_1: \theta \neq \theta_0$

# Types of error

| Decision based on sample | | Truth about the population: $H_0$ true | Truth about the population: $H_a$ true |
|---|---|---|---|
| | Reject $H_0$ | Type I error | Correct decision |
| | Accept $H_0$ | Correct decision | Type II error |

# Hypothesis

- Consider an example that a person is caught and the judgement is to be passed by the Judge whether the person caught is guilty or not.
- Hypothesis test,

  $H_0$: person caught is not guilty

  $H_1$: person caught is guilty

# Types of error

| Decision based on sample | Truth about the population: $H_0$ true | Truth about the population: $H_a$ true |
|---|---|---|
| Reject $H_0$ | Type I error<br>$\alpha$ | Correct decision<br>$1 - \alpha$ |
| Accept $H_0$ | Correct decision<br>$1 - \beta$ | Type II error<br>$\beta$ |

# Hypothesis

- ***Type I error, α***: Judge pronounces that a person caught is guilty when he is really not guilty.
- ***Type II error,* β**: Judge pronounces that a person caught is not guilty when he is really guilty.
- **1-β, power of test:** Judge pronounces that a person caught is guilty when he is really guilty.

# Test Statistics

- Assume a situation where we have sample ($x_1$, $x_2$,.....,$x_n$) of size n and there is a suspicion that this sample is from a normal population with mean μ and variance $\sigma^2$
- If $\hat{\theta}$ is the sample mean obtained from the sample of size n such that $\hat{\theta}=\bar{X}$ , then under assumption $\mu= \mu_0$, by central limit theorem

$$Z=\frac{|\bar{X}-\mu|}{\sigma/\sqrt{n}}=\frac{\left|\bar{\hat{\theta}}-\mu_0\right|}{\sigma/\sqrt{n}}$$

is called test statistics

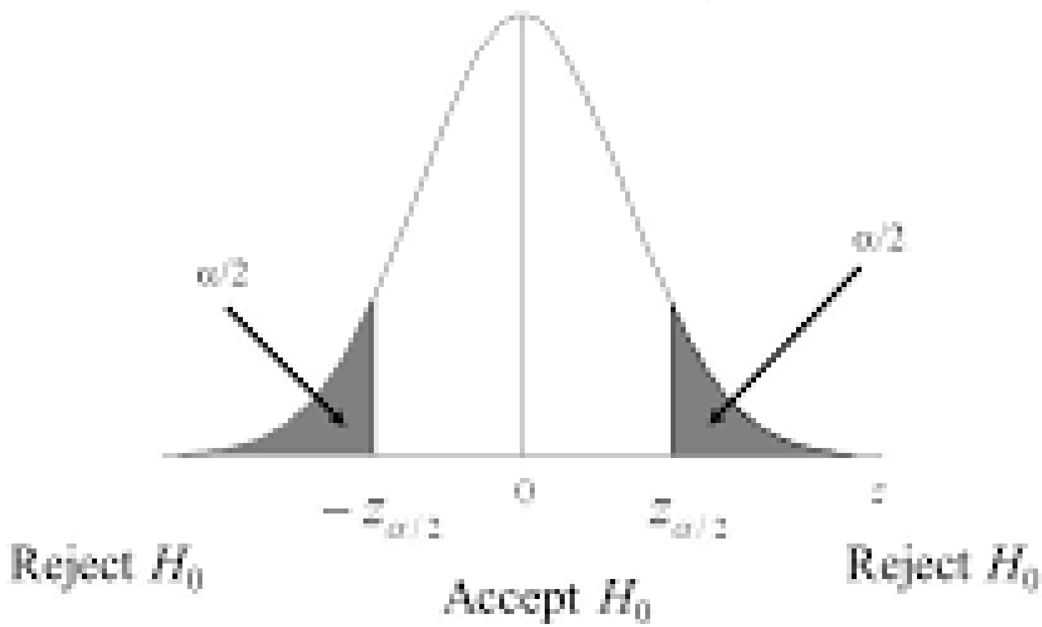
# Test Types

- Two tailed test : rejection area are on both sides of acceptance area

- One tailed test: rejection are on one side of acceptance area

# Two sided/tailed test

***The Critical region:***




$$P[\text{Accept } H_0 \text{ when true}] = P[-z_{\alpha/2} \le z \le z_{\alpha/2}] = 1-\alpha$$

$$P[\text{Reject } H_0 \text{ when true}] = P[z < -z_{\alpha/2} \text{ or } z > z_{\alpha/2}] = \alpha$$

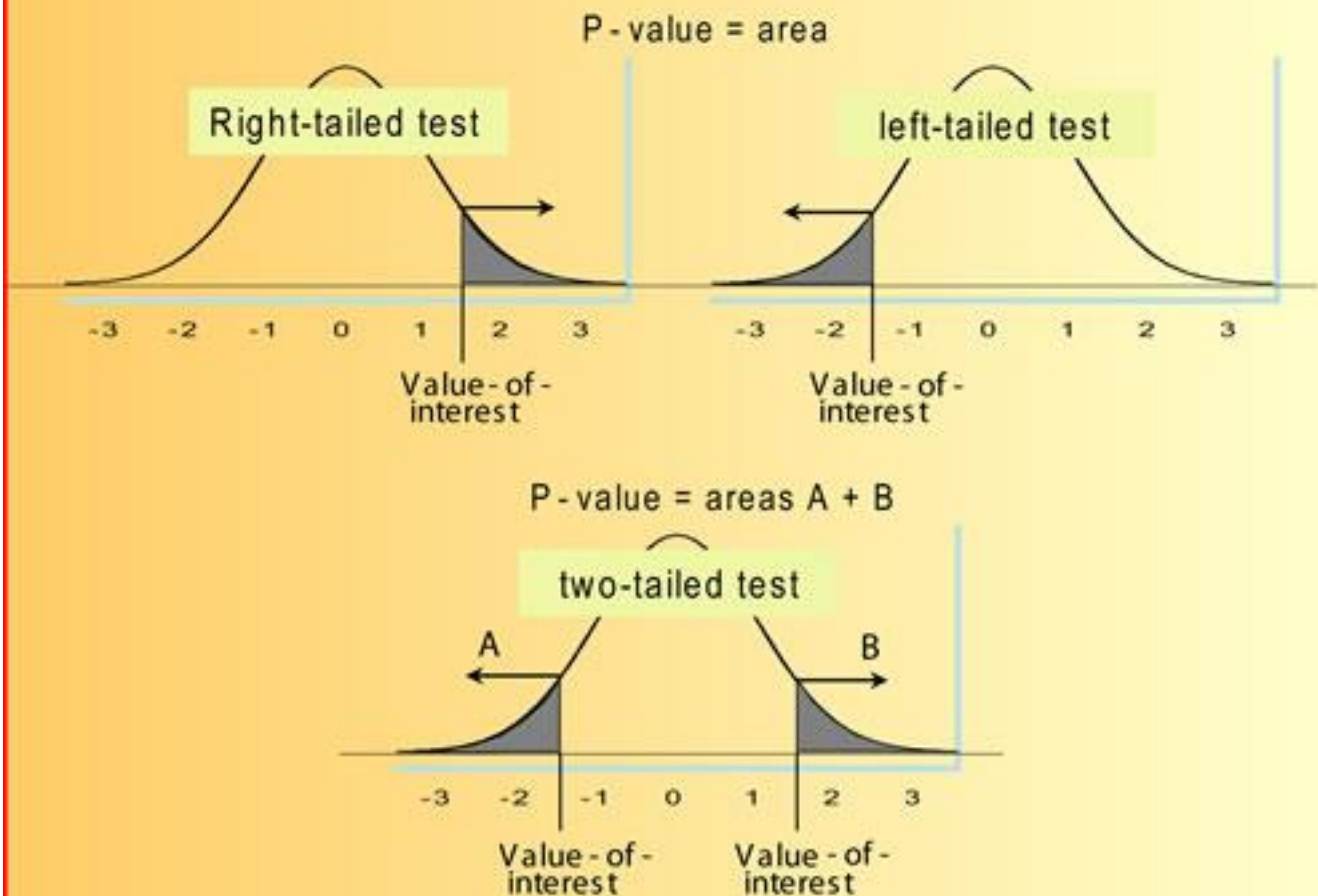
# Two tailed

- **Acceptance region**: $-z_{\alpha/2} \leq Z \leq + z_{\alpha/2}$
- **Critical regions** : $(Z< -z_{\alpha/2})$ and $(Z > + z_{\alpha/2})$
- **Critical values**: $-z_{\alpha/2}$ and $+ z_{\alpha/2}$
- $P(Z< -z_{\alpha/2}) + P(Z > + z_{\alpha/2}) = \alpha$
- Hypothesis test is:

*Null hypothesis* $H_0: \theta = \theta_0$

*Alternate Hypothesis* $H_1: \theta \neq \theta_0$

$$H_1: \theta \neq \theta_0 \Rightarrow \theta < \theta_0 \text{ or } \theta > \theta_0$$

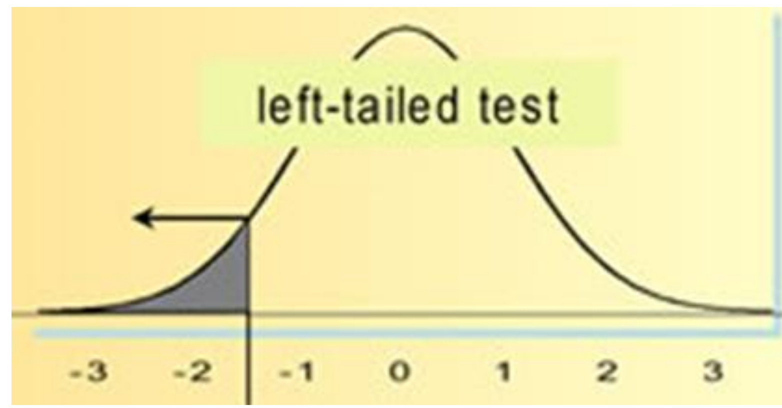

P - value = area
Right-tailed test
left-tailed test
-3
-2
-1
0
1
2
3
Value - of - interest
P - value = areas A + B
two-tailed test
A
B

# Left Tail Test

- **Test statistics**: $Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} = \frac{\widehat{\theta} - \mu_0}{\sigma/\sqrt{n}} = z_1$
- **Acceptance region**: $-z_{\alpha/2} \leq Z$
- **Critical regions** : $(Z < -z_{\alpha/2})$
- **Critical values**: $-z_{\alpha/2}$
- $P(Z < -z_{\alpha/2}) = \alpha$
- Hypothesis test is:

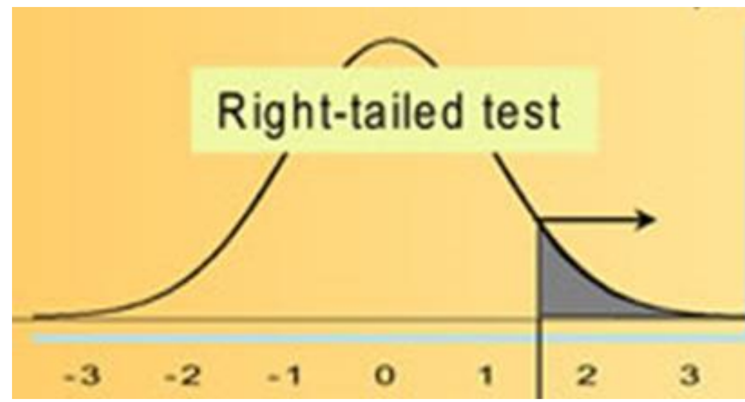
*Null hypothesis* $H_0: \theta = \theta_0$

*Alternate Hypothesis* $H_1: \theta < \theta_0$



# Right Tailed Test

- **Test statistics**: $Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} = \frac{\hat{\theta} - \mu_0}{\sigma/\sqrt{n}} = z_2$
- **Acceptance region**: $Z \leq + z_{\alpha/2}$
- **Critical regions** : $(Z > + z_{\alpha/2})$
- **Critical values**: $+ z_{\alpha/2}$
- $P(Z > + z_{\alpha/2}) = \alpha$
- Hypothesis test is:



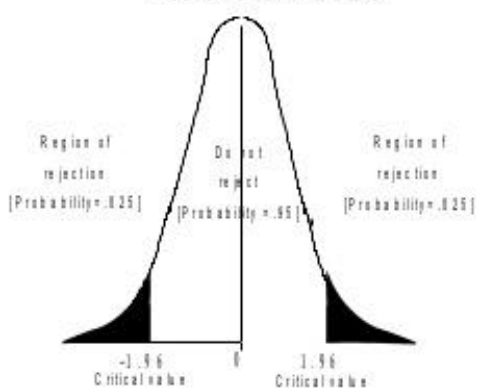

*Null hypothesis* $H_0: \theta = \theta_0$

*Alternate Hypothesis* $H_1: \theta > \theta_0$

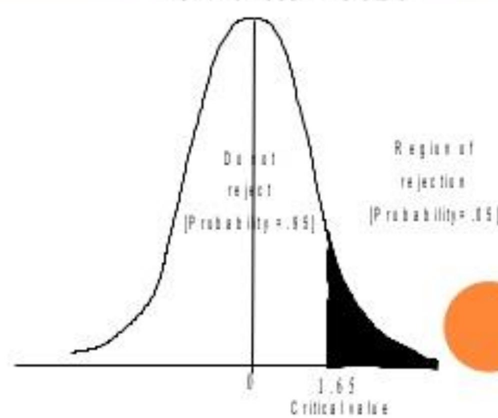
# Values of Normal distribution for confidence interval

| Level of significant (α) Type 1 error | Confidence interval | Z value for two tail test | Z value for one tail test |
|---|---|---|---|
| 0.01 | 99% | p= 0.005<br>z= 2.58 | p=0.01<br>Z=2.33 |
| 0.05 | 95% | p=0.025<br>z = 1.96 | p=0.05<br>Z=1.65 |
| 0.10 | 90% | p=0.05<br>z = 1.65 | p=0.10<br>Z=1.28 |

**Two tail test**




**One tail test**



# One tailed Test

We can also test the one-sided hypothesis

$$H_0 : \mu = \mu_0 \quad (\text{or } \mu \geq \mu_0) \qquad \text{versus} \quad H_1 : \mu < \mu_0$$

at significance level $\alpha$ by

$$\text{accepting} \quad H_0 \quad \text{if} \quad \frac{\sqrt{n}}{\sigma}(\overline{X} - \mu_0) \geq -z_\alpha$$

$$\text{rejecting} \quad H_0 \quad \text{otherwise}$$

or

$$\text{accept} \quad H_0 \quad \text{if} \quad \frac{\sqrt{n}}{\sigma}(\overline{X} - \mu_0) \leq z_\alpha$$

$$\text{reject} \quad H_0 \quad \text{if} \quad \frac{\sqrt{n}}{\sigma}(\overline{X} - \mu_0) > z_\alpha$$

# Testing the of mean of a normal population

- Hypothesis testing for population mean μ when population *variance $\sigma^2$ is known*
- Hypothesis testing for population mean μ when population *variance $\sigma^2$ is unknown*

# Hypothesis testing for population mean μ when population variance **<u>$\sigma^2$ is known</u>**

# Steps for Hypothesis testing (Normal Test, Variance known)

- **Step 1:** Propose the Null hypothesis $H_0$
- **Step 2:** Propose the alternative hypothesis $H_1$
- **Step 3:** Assume the level of significance such as $\alpha$=0.05 (5%) and $\alpha$=0.01 (1%) and obtain the critical values
  - $-z\alpha/2$ and $+z\alpha/2$ for Two tailed test
  - $-z\alpha/2$ for Left Tail Test
  - $+z\alpha/2$ for Right Tailed Test

# Steps for Hypothesis testing (Normal Test)

- **Step 4:** Compute Z value with proper substitution of values in Test statistics $Z=\frac{\bar{X}-\mu}{\sigma/\sqrt{n}}= \frac{\hat{\theta}-\mu_0}{\sigma/\sqrt{n}}$
- **Step 5** : Inference

➢ For 2 tailed test: if calculated Z value lies in the critical region i.e. ($Z< -z_{\alpha/2}$) and ($Z > + z_{\alpha/2}$) then reject the Null Hypothesis that $H_0$: $\theta= \theta_0$, which means that the sample from which the statics $\hat{\theta}$ is obtained is not from the population whose parameter is $\theta= \theta_0$

# Problem

**EXAMPLE 8.3a** It is known that if a signal of value $\mu$ is sent from location A, then the value received at location B is normally distributed with mean $\mu$ and standard deviation 2. That is, the random noise added to the signal is an $N(0, 4)$ random variable. There is reason for the people at location B to suspect that the signal value $\mu = 8$ will be sent today. Test this hypothesis if the same signal value is independently sent five times and the average value received at location B is $\overline{X} = 9.5$.

Suppose we are testing at the 5 percent level of significance.

# Solution

- Given: $\sigma=2$,

  $\mu_0 = 8$,

  $\alpha = 0.05$, $\overline{X} = 9.5$

  $n=5$

*For two sided test*

$$\frac{\sqrt{n}}{\sigma}|\overline{X} - \mu_0| = \frac{\sqrt{5}}{2}(1.5) = 1.68$$

*Null hypothesis* $H_0: \theta = \theta_0$

*Alternate Hypothesis* $H_1: \theta \neq \theta_0$

Since this value is less than $z_{.025} = 1.96$, the hypothesis is accepted.

# Choosing level of significance

- However, that if a less stringent significance level were chosen — say $\alpha = 0.1$— then the null hypothesis would have been rejected.
- This follows since $z_{.05}$ =1.645, which is less than 1.68.
- Hence, if we would have chosen a test that had a 10 percent chance of rejecting $H_0$ when $H_0$ was true, then the null hypothesis would have been rejected.

# Choosing level of significance

- The “correct” level of significance to use in a given situation depends on the individual circumstances involved in that situation.
- For instance, if rejecting a null hypothesis $H_0$ would result in large costs that would thus be lost if $H_0$ were indeed true, then we might choose a significance level of .05 or .01.

# One tailed Test

We can also test the one-sided hypothesis

$$H_0 : \mu = \mu_0 \quad (\text{or } \mu \geq \mu_0) \qquad \text{versus} \quad H_1 : \mu < \mu_0$$

at significance level $\alpha$ by

$$\text{accepting} \quad H_0 \quad \text{if} \quad \frac{\sqrt{n}}{\sigma}(\overline{X} - \mu_0) \geq -z_\alpha$$

$$\text{rejecting} \quad H_0 \quad \text{otherwise}$$

or

$$\text{accept} \quad H_0 \quad \text{if} \quad \frac{\sqrt{n}}{\sigma}(\overline{X} - \mu_0) \leq z_\alpha$$

$$\text{reject} \quad H_0 \quad \text{if} \quad \frac{\sqrt{n}}{\sigma}(\overline{X} - \mu_0) > z_\alpha$$

# Problem

**EXAMPLE 8.3f** All cigarettes presently on the market have an average nicotine content of at least 1.6 mg per cigarette. A firm that produces cigarettes claims that it has discovered a new way to cure tobacco leaves that will result in the average nicotine content of a cigarette being less than 1.6 mg. To test this claim, a sample of 20 of the firm's cigarettes were analyzed. If it is known that the standard deviation of a cigarette's nicotine content is .8 mg, what conclusions can be drawn, at the 5 percent level of significance, if the average nicotine content of the 20 cigarettes is 1.54?

# Solution

- The standard reading always becomes Null hypothesis and the claim as the alternate hypothesis

$$H_0 : \mu \geq 1.6 \qquad \text{versus} \qquad H_1 : \mu < 1.6$$

- The value of the Test Statistic(TS) is

$$\sqrt{n}(\overline{X} - \mu_0)/\sigma = \sqrt{20}(1.54 - 1.6)/.8 = -.336$$

- For one sided test Reject $H_0$ if $TS < -z_\alpha$
- $-z_\alpha = -z_{0.05} = -1.65$

# Values of Normal distribution for confidence interval

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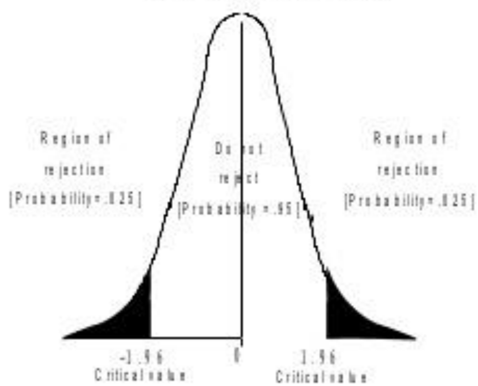
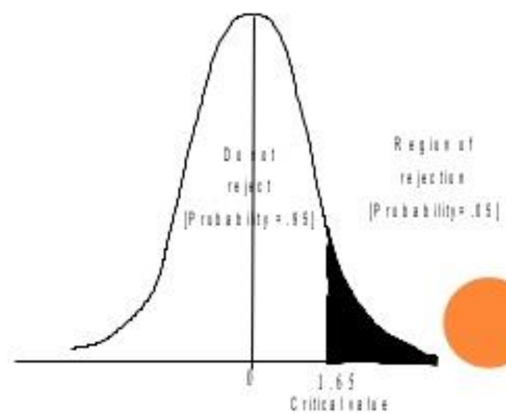
| 0.10 | 90% | p=0.05<br>z = 1.65 | p=0.10<br>Z=1.28 |

Two tail test




One tail test

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# Solution

- As $TS > -z_{\alpha}$ we accept the Null hypothesis
- Hence the foregoing data do not enable us to reject, at the .05 percent level of significance, the hypothesis that the mean nicotine content exceeds 1.6mg.

**TABLE 8.1** *$X_1, \ldots, X_n$ Is a Sample from a $\mathcal{N}(\mu, \sigma^2)$*

*Population $\sigma^2$ Is Known, $\overline{X} = \sum_{i=1}^{n} X_i/n$*

| $H_0$ | $H_1$ | Test Statistic $TS$ | Significance Level $\alpha$ Test | $p$-Value if $TS = t$ |
|---|---|---|---|---|
| $\mu = \mu_0$ | $\mu \neq \mu_0$ | $\sqrt{n}(\overline{X} - \mu_0)/\sigma$ | Reject if $\|TS\| > z_{\alpha/2}$ | $2P\{Z \geq \|t\|\}$ |
| $\mu \leq \mu_0$ | $\mu > \mu_0$ | $\sqrt{n}(\overline{X} - \mu_0)/\sigma$ | Reject if $TS > z_\alpha$ | $P\{Z \geq t\}$ |
| $\mu \geq \mu_0$ | $\mu < \mu_0$ | $\sqrt{n}(\overline{X} - \mu_0)/\sigma$ | Reject if $TS < -z_\alpha$ | $P\{Z \leq t\}$ |

*$Z$ is a standard normal random variable.*

# Hypothesis testing for population mean μ when population variance

# **$\sigma^2$ is unknown**

# Case of Unknown Variance: The *t*-Test

- When population variance $\sigma^2$ is no longer known and the only unknown parameter is the population mean, it is estimate by

$$S^2 = \frac{\sum_{i=1}^{n} (X_i - \overline{X})^2}{n-1}$$

- The statistic *T is* defined by

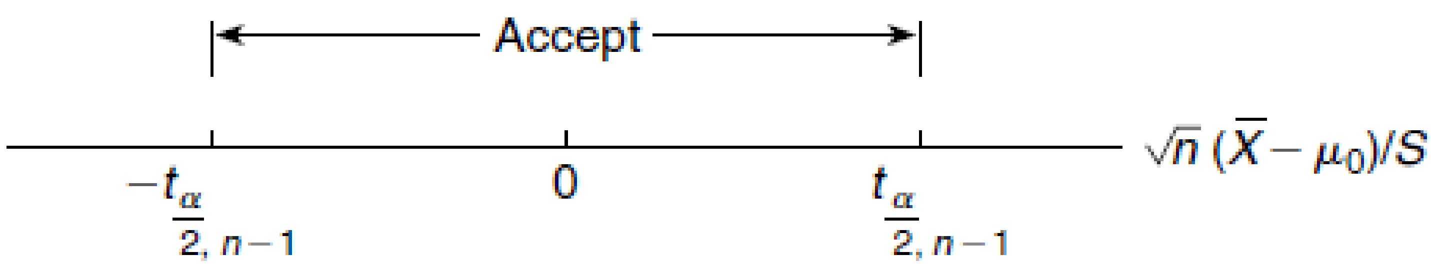
$$T = \frac{\sqrt{n}(\overline{X} - \mu_0)}{S}$$

# *Two-sided t-test*

- When $\sigma^2$ is unknown

$$\text{accept} \quad H_0 \quad \text{if} \quad \left| \frac{\sqrt{n}(\overline{X} - \mu_0)}{S} \right| \leq t_{\alpha/2,\, n-1}$$

$$\text{reject} \quad H_0 \quad \text{if} \quad \left| \frac{\sqrt{n}(\overline{X} - \mu_0)}{S} \right| > t_{\alpha/2,\, n-1}$$



# One sided test

We can use a one-sided $t$-test to test the hypothesis

$$H_0 : \mu = \mu_0 \qquad (\text{or } H_0 : \mu \leq \mu_0)$$

$$H_1 : \mu > \mu_0$$

The significance level $\alpha$ test is to

$$\text{accept} \quad H_0 \quad \text{if} \quad \frac{\sqrt{n}(\overline{X} - \mu_0)}{S} \leq t_{\alpha,\, n-1}$$

$$\text{reject} \quad H_0 \quad \text{if} \quad \frac{\sqrt{n}(\overline{X} - \mu_0)}{S} > t_{\alpha,\, n-1}$$

# One sided test

The significance level $\alpha$ test of

$$H_0 : \mu = \mu_0 \qquad (\text{or } H_0 : \mu \geq \mu_0)$$

versus the alternative

$$H_1 : \mu < \mu_0$$

is to

$$\text{accept} \quad H_0 \quad \text{if} \quad \frac{\sqrt{n}(\overline{X} - \mu_0)}{S} \geq -t_{\alpha,\, n-1}$$

$$\text{reject} \quad H_0 \quad \text{if} \quad \frac{\sqrt{n}(\overline{X} - \mu_0)}{S} < -t_{\alpha,\, n-1}$$

# Problem

**EXAMPLE 8.3h** A public health official claims that the mean home water use is 350 gallons a day. To verify this claim, a study of 20 randomly selected homes was instigated with the result that the average daily water uses of these 20 homes were as follows:

| | | | |
|---|---|---|---|
| 340 | 344 | 362 | 375 |
| 356 | 386 | 354 | 364 |
| 332 | 402 | 340 | 355 |
| 362 | 322 | 372 | 324 |
| 318 | 360 | 338 | 370 |

Do the data contradict the official's claim?

Suppose the claim is to be tested for 10% level of significance

# Solution

To determine if the data contradict the official's claim, we need to test

$$H_0 : \mu = 350 \qquad \text{versus} \qquad H_1 : \mu \neq 350$$

$$\overline{X} = 353.8, \qquad S = 21.8478$$

the value of the test statistic is $\quad T = \dfrac{\sqrt{20}(3.8)}{21.8478} = .7778$

Because this is less than $t_{.05,19} = 1.730$, the null hypothesis is accepted at the 10 percent level of significance. Indeed, the $p$-value of the test data is

# Problem

**EXAMPLE 8.3i** The manufacturer of a new fiberglass tire claims that its average life will be at least 40,000 miles. To verify this claim a sample of 12 tires is tested, with their lifetimes (in 1,000s of miles) being as follows:

| **Tire** | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Life** | 36.1 | 40.2 | 33.8 | 38.5 | 42 | 35.8 | 37 | 41 | 36.8 | 37.2 | 33 | 36 |

Test the manufacturer's claim at the 5 percent level of significance.

# Solution

- To determine whether the foregoing data are consistent with the hypothesis that the mean life is at least 40,000 miles, we will test

$$H_0 : \mu \geq 40{,}000 \qquad \text{versus} \qquad H_1 : \mu < 40{,}000$$

$$\overline{X} = 37.2833, \qquad S = 2.7319$$

$$T = \frac{\sqrt{12}(37.2833 - 40)}{2.7319} = -3.4448$$

Since this is less than $-t_{.05,11} = -1.796$, the null hypothesis is rejected at the 5 percent

# Testing the equality of means of two normal populations

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# Testing the equality of means of two normal populations

- Case of Known Variances
- Case of Unknown and equal Variances

# Case of Known Variances

# Case of Known Variances

- Suppose that $X_1, . . . , X_n$ and $Y_1, . . . , Y_m$ are independent samples from normal populations having unknown population means $\mu_x$ and $\mu_y$ but known population variances $\sigma^2{}_x$ and $\sigma^2{}_y$
- Let us consider the problem of testing the hypothesis

# Case of Known Variances

- The hypothesis can be stated as

$$H_0 : \mu_x = \mu_y$$

versus the alternative

$$H_1 : \mu_x \neq \mu_y$$

- Since $\bar{X}$ is an estimate of $\mu_x$ and $\bar{Y}$ of $\mu_y$ , it follows that $\bar{X} - \bar{Y}$ can be used to estimate

$\mu_x - \mu_y$ .

- Hence, the null hypothesis can be written as

$$H_0 : \mu_x - \mu_y = 0$$

# **Case of Known Variances:** Two sided test

- Test statistics is $(\overline{X}-\overline{Y})\big/\sqrt{\sigma_x^2/n+\sigma_y^2/m}.$
- We obtain two sided test for the significance level $\alpha$ test of $H_0 : \mu_x = \mu_y$ versus $H_1 : \mu_x \neq \mu_y$ *as*

$$\text{accept} \quad H_0 \quad \text{if} \quad \frac{|\overline{X}-\overline{Y}|}{\sqrt{\sigma_x^2/n+\sigma_y^2/m}} \le z_{\alpha/2}$$

$$\text{reject} \quad H_0 \quad \text{if} \quad \frac{|\overline{X}-\overline{Y}|}{\sqrt{\sigma_x^2/n+\sigma_y^2/m}} \ge z_{\alpha/2}$$

# **Case of Known Variances:** One sided test

- We obtain one sided test for the significance level $\alpha$ test of $H_0 : \mu_x \leq \mu_y$ versus $H_1 : \mu_x > \mu_y$ *as*

$$\text{accept} \quad H_0 \quad \text{if} \quad \overline{X} - \overline{Y} \leq z_\alpha \sqrt{\frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{m}}$$

$$\text{reject} \quad H_0 \quad \text{if} \quad \overline{X} - \overline{Y} > z_\alpha \sqrt{\frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{m}}$$

# Case of Unknown and equal Variances

# Case of Unknown Variances

- Suppose again that $X_1, \ldots, X_n$ and $Y_1, \ldots, Y_m$ are independent samples from normal populations having respective population parameters $(\mu_x, \sigma^2_x)$ and $(\mu_y, \sigma^2_y)$, but now suppose that all four parameters are unknown.
- We will once again consider a test of

$$H_0 : \mu_x = \mu_y \text{ versus } H_1 : \mu_x \neq \mu_y$$

# Case of Unknown and equal Variances

- To determine a significance level $\alpha$ test of $H_0$ we will need to make the additional assumption that the unknown variances $\sigma^2_x$ and $\sigma^2_y$ are equal.
- Let $\sigma^2$ denote their value—that is,

$$\sigma^2 = \sigma^2_x = \sigma^2_y$$

- The sample variance of the two sample are given as

$$S_x^2 = \frac{\sum_{i=1}^{n}(X_i - \overline{X})^2}{n-1} \qquad S_y^2 = \frac{\sum_{i=1}^{m}(Y_i - \overline{Y})^2}{m-1}$$

# **Case of Unknown and equal Variances:** Two sided test

- When $H_0$ is true, and so $\mu_x - \mu_y = 0$, the test statistic has a $t$ -distribution with $n + m - 2$ degrees of freedom and is given as

$$T \equiv \frac{\overline{X} - \overline{Y}}{\sqrt{S_p^2(1/n + 1/m)}}$$

where $S^2{}_p$ , the *pooled* estimator of the common variance $\sigma^2$, is given by

$$S_p^2 = \frac{(n-1)S_x^2 + (m-1)S_y^2}{n+m-2}$$

# **Case of Unknown and equal Variances:**
Two sided test

- It follows that we can test the hypothesis that $\mu_x = \mu_y$ as follows:

$$\begin{array}{llll} \text{accept} & H_0 & \text{if} & |T| \leq t_{\alpha/2,\, n+m-2} \\ \text{reject} & H_0 & \text{if} & |T| > t_{\alpha/2,\, n+m-2} \end{array}$$

- where $t_{\alpha/2,\, n+m-2}$ is the 100 $\alpha/2$ percentile point of a $t$-random variable with $n + m - 2$degrees of freedom

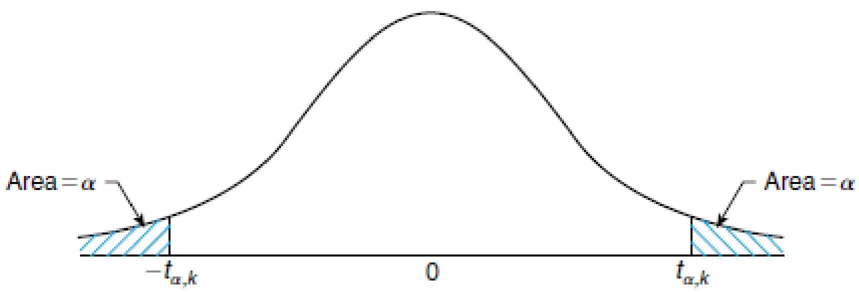


FIGURE 8.5 *Density of a t-random variable with k degrees of freedom.*

# Case of Unknown and equal Variances:
## One sided test

- In testing the one-sided hypothesis

$$H_0 : \mu_x \le \mu_y \text{ versus } H_1 : \mu_x > \mu_y$$

- Then $H_0$ will be rejected at large values of T
- Thus the significance level $\alpha$ test is to

$$\begin{array}{llll} \text{accept} & H_0 & \text{if} & T < t_{\alpha, n+m-2} \\ \text{reject} & H_0 & \text{if} & T \ge t_{\alpha, n+m-2} \end{array}$$

**EXAMPLE 8.4b** Twenty-two volunteers at a cold research institute caught a cold after having been exposed to various cold viruses. A random selection of 10 of these volunteers was given tablets containing 1 gram of vitamin C. These tablets were taken four times a day. The control group consisting of the other 12 volunteers was given placebo tablets that looked and tasted exactly the same as the vitamin C tablets. This was continued for each volunteer until a doctor, who did not know if the volunteer was receiving the vitamin C or the placebo tablets, decided that the volunteer was no longer suffering from the cold. The length of time the cold lasted was then recorded.

At the end of this experiment, the following data resulted.

| Treated with Vitamin C | Treated with Placebo |
|---|---|
| 5.5 | 6.5 |
| 6.0 | 6.0 |
| 7.0 | 8.5 |
| 6.0 | 7.0 |
| 7.5 | 6.5 |
| 6.0 | 8.0 |
| 7.5 | 7.5 |
| 5.5 | 6.5 |
| 7.0 | 7.5 |
| 6.5 | 6.0 |
| | 8.5 |
| | 7.0 |

Do the data listed prove that taking 4 grams daily of vitamin C reduces the mean length of time a cold lasts? At what level of significance?

# Solution

- *Let* $\mu_c$ be the mean time a cold lasts when the vitamin C tablets are taken and $\mu_p$ be the mean time when the placebo is taken.
- The hypothesis can be given as

*Null hypothesis* $H_0: \mu_c \le \mu_p$

*Alternate Hypothesis* $H_1: \mu_c > \mu_p$

$$\text{accept} \quad H_0 \quad \text{if} \quad T < t_{\alpha, n+m-2}$$
$$\text{reject} \quad H_0 \quad \text{if} \quad T \geq t_{\alpha, n+m-2}$$

# Solution

$$\overline{X} = 6.450, \qquad \overline{Y} = 7.125$$

$$S_x^2 = .581, \qquad S_y^2 = .778$$

$$S_p^2 = \frac{9}{20}S_x^2 + \frac{11}{20}S_y^2 = .689$$

and the value of the test statistic is

$$TS = \frac{-.675}{\sqrt{.689(1/10 + 1/12)}} = -1.90$$

Since $t_{0.5,20} = 1.725$,

That is, at the 5 percent level of significance the evidence is significant in establishing that vitamin C reduces the mean time that a cold persists. ■

**TABLE 8.4** *$X_1, \ldots, X_n$ Is a Sample from a $\mathcal{N}(\mu_1, \sigma_1^2)$ Population; $Y_1, \ldots, Y_m$ Is a Sample from a $\mathcal{N}(\mu_2, \sigma_2^2)$ Population*

*The Two Population Samples Are Independent*

*to Test*

$H_0 : \mu_1 = \mu_2$ versus $H_0 : \mu_1 \neq \mu_2$

| Assumption | Test Statistic $TS$ | Significance Level $\alpha$ Test | $p$-Value if $TS = t$ |
|---|---|---|---|
| $\sigma_1, \sigma_2$ known | $\dfrac{\overline{X}-\overline{Y}}{\sqrt{\sigma_1^2/n+\sigma_2^2/m}}$ | Reject if $\lvert TS \rvert > z_{\alpha/2}$ | $2P\{Z \geq \lvert t \rvert\}$ |
| $\sigma_1 = \sigma_2$ | $\dfrac{\overline{X}-\overline{Y}}{\sqrt{\frac{(n-1)S_1^2+(m-1)S_2^2}{n+m-2}}\sqrt{1/n+1/m}}$ | Reject if $\lvert TS \rvert > t_{\alpha/2,\, n+m-2}$ | $2P\{T_{n+m-2} \geq \lvert t \rvert\}$ |

| Testing of | Population Variance $\sigma^2$ | Test Statistic | $H_0: \mu = \mu_0$ <br> $H_1: \mu \neq \mu_0$ | $H_0 : \mu \leq \mu_0$ <br> $H_1 : \mu > \mu_0$ | $H_0 : \mu \geq \mu_0$ <br> $H_1 : \mu < \mu_0$ |
|---|---|---|---|---|---|
| Population mean $\mu$ | known | $Z=\frac{\bar{X}-\mu}{\sigma/\sqrt{n}}$ | accept $H_0$ if $\frac{\sqrt{n}}{\sigma}\lvert\bar{X}-\mu_0\rvert \leq z_{\alpha/2}$ <br> reject $H_0$ if $\frac{\sqrt{n}}{\sigma}\lvert\bar{X}-\mu_0\rvert > z_{\alpha/2}$ | accept $H_0$ if $\frac{\sqrt{n}}{\sigma}(\bar{X}-\mu_0) \leq z_{\alpha}$ <br> reject $H_0$ if $\frac{\sqrt{n}}{\sigma}(\bar{X}-\mu_0) > z_{\alpha}$ | accepting $H_0$ if $\frac{\sqrt{n}}{\sigma}(\bar{X}-\mu_0) \geq -z_{\alpha}$ <br> rejecting $H_0$ otherwise |
| | unknown | $T=\frac{\sqrt{n}(\bar{X}-\mu_0)}{S}$ | accept $H_0$ if $\left\lvert\frac{\sqrt{n}(\bar{X}-\mu_0)}{S}\right\rvert \leq t_{\alpha/2,\,n-1}$ <br> reject $H_0$ if $\left\lvert\frac{\sqrt{n}(\bar{X}-\mu_0)}{S}\right\rvert > t_{\alpha/2,\,n-1}$ | accept $H_0$ if $\frac{\sqrt{n}(\bar{X}-\mu_0)}{S} \leq t_{\alpha,\,n-1}$ <br> reject $H_0$ if $\frac{\sqrt{n}(\bar{X}-\mu_0)}{S} > t_{\alpha,\,n-1}$ | accept $H_0$ if $\frac{\sqrt{n}(\bar{X}-\mu_0)}{S} \geq -t_{\alpha,\,n-1}$ <br> reject $H_0$ if $\frac{\sqrt{n}(\bar{X}-\mu_0)}{S} < -t_{\alpha,\,n-1}$ |
| Equality of means of two normal populations | Known | $(\bar{X}-\bar{Y})/\sqrt{\sigma_x^2/n+\sigma_y^2/m}$ | accept $H_0$ if $\frac{\lvert\bar{X}-\bar{Y}\rvert}{\sqrt{\sigma_x^2/n+\sigma_y^2/m}} \leq z_{\alpha/2}$ <br> reject $H_0$ if $\frac{\lvert\bar{X}-\bar{Y}\rvert}{\sqrt{\sigma_x^2/n+\sigma_y^2/m}} \geq z_{\alpha/2}$ | accept $H_0$ if $\bar{X}-\bar{Y} \leq z_{\alpha}\sqrt{\frac{\sigma_x^2}{n}+\frac{\sigma_y^2}{m}}$ <br> reject $H_0$ if $\bar{X}-\bar{Y} > z_{\alpha}\sqrt{\frac{\sigma_x^2}{n}+\frac{\sigma_y^2}{m}}$ | accept $H_0$ if $\bar{X}-\bar{Y} \geq -z_{\alpha}\sqrt{\frac{\sigma_x^2}{n}+\frac{\sigma_y^2}{m}}$ <br> reject $H_0$ if $\bar{X}-\bar{Y} < -z_{\alpha}\sqrt{\frac{\sigma_x^2}{n}+\frac{\sigma_y^2}{m}}$ |
| | Unknown and Equal | $T=\frac{\bar{X}-\bar{Y}}{\sqrt{S_p^2(1/n+1/m)}}$ | accept $H_0$ if $\lvert T\rvert \leq t_{\alpha/2,\,n+m-2}$ <br> reject $H_0$ if $\lvert T\rvert > t_{\alpha/2,\,n+m-2}$ | accept $H_0$ if $T < t_{\alpha,\,n+m-2}$ <br> reject $H_0$ if $T \geq t_{\alpha,\,n+m-2}$ | accept $H_0$ if $T \geq -t_{\alpha,\,n+m-2}$ <br> reject $H_0$ if $T < -t_{\alpha,\,n+m-2}$ |

$$S^2=\frac{\sum_{i=1}^{n}(X_i-\bar{X})^2}{n-1} \qquad S_p^2=\frac{(n-1)S_x^2+(m-1)S_y^2}{n+m-2}$$

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